JUHO JEON

Hong Kong jjeonaa@connect.ust.hk

EDUCATION

The Hong Kong University of Science and Technology (HKUST) Sep 2024 – Dec 2026 (Expected)
Master of Philosophy (MPhil) in Electronic and Computer Engineering *Hong Kong*

- GPA: 3.717 / 4.30 | Advisor: Prof. Patrick Chik Yue
- Teaching Assistant, ELEC3400 – Integrated Circuit Designs and Systems (Spring 2025)

The Hong Kong University of Science and Technology (HKUST) Sep 2017 – Jun 2024
Bachelor of Engineering in Electronic Engineering; Minor in Information Technology *Hong Kong*

- GPA: 3.09 / 4.30
- Academic leave: Sep 2019 – Aug 2021 (mandatory military service, Republic of Korea); Feb 2022 – Feb 2023 (full-time internship, Deloitte Advisory)

PUBLICATIONS

- J. Jeon and C. P. Yue, *"Digitally Reconfigurable Resolution Multi-Bit-per-Cycle SAR ADC: Design Trade-off Considerations,"* **IEEE Region 10 Conference (TENCON) 2026**, accepted.

RESEARCH EXPERIENCE

28nm CMOS Reconfigurable Multi-Resolution SAR ADC — MPhil Thesis Sep 2024 – Present
Advisor: Prof. Patrick Chik Yue | Post-Layout Simulation

- Designed a digitally reconfigurable SAR ADC supporting five resolution modes (5b–9b) via a 1-then-2-bit-per-cycle conversion scheme, with reconfigurability implemented entirely through digital logic — no switches added to the analog signal path.
- Achieved 51.84 dB SNDR, 8.31 ENOB, and 66.08 dB SFDR at 1.4 GS/s (0.9 V supply, 9b mode), corresponding to a 23.8 fJ/conversion-step Walden FoM and 160 dB Schreier FoM, verified using post-layout extracted netlists for the DAC and comparator.
- Designed an enable-gated DAC switching architecture and mode-aware decoder allowing seamless mode transitions via a 4-bit control word with no analog reconfiguration.
- Quantified the high-resolution-limited power trade-off (~9× FoM degradation from 9b to 5b mode) arising from sharing a single DAC/comparator across all resolution modes.
- Results accepted for presentation at IEEE TENCON 2026.

180nm CMOS 10-bit SAR ADC — Final Year Thesis Sep 2023 – May 2024
Advisor: Prof. Zhang Yihan | Schematic-Level Simulation, TSMC 180nm | First Runner-Up, ECE Best FYP Award

- Designed and independently verified a 10-bit monotonic-switching SAR ADC (capacitive DAC, bootstrapped sampling switch, dynamic comparator, asynchronous SAR logic) in Cadence Virtuoso, achieving 9.09 ENOB and 1.92 mW power consumption at 2 MS/s.

COURSE PROJECTS

28GHz Differential LNA Design Spring 2026
RFIC Design Coursework (Prof. Patrick Chik Yue) | 22nm TSMC CMOS

- Designed a 28GHz differential cascode LNA with inductive source/gate degeneration, achieving 17.4 dB gain, 1.74 dB NF, –14.1 dB S11, and 9.16 dBm IIP3 at 22.8 mW.

Gain-Boosted Folded-Cascode Fully-Differential OTA Fall 2024
Advanced Analog IC Design Coursework (Prof. Howard Luong) | 90nm TSMC CMOS

- Designed a gain-boosted folded-cascode OTA with resistive CMFB, achieving 81.6 dB gain, 604 MHz UGF, 61.5° phase margin, 103 dB CMRR, and –66 dB THD at 1.93 mW (±1 V supply).

Continuous-Time Gm-C Channel-Selection Filter for UWB Fall 2024
Advanced Analog IC Design Coursework (Prof. Howard Luong) | 90nm TSMC CMOS

- Designed a Gm-C UWB bandpass filter (35 kHz–259 MHz) from passive Chebyshev-I prototypes through transistor-level implementation, meeting passband gain/ripple, corner-frequency, and adjacent-channel specs.

DC-DC Buck Converter with PWM Voltage-Mode Control Fall 2024
Power Management IC Design Coursework (Prof. Wing Hung Ki) | 180nm TSMC CMOS

- Designed a 3.3V-to-1.8V, 1MHz synchronous buck converter achieving 86.2% peak efficiency and 45.2 mV ripple across 200–800 mA CCM load.
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WORK EXPERIENCE

Deloitte Advisory Feb 2022 – Dec 2022
Cyber Security Intern *Hong Kong*

- Developed a demo web application implementing Identity and Access Management (IAM) features using Okta, Salesforce, and Hong Kong iAMSmart.
- Designed, tested, and implemented IAM workflows for client engagements.

Gense Technologies Dec 2021 – Jan 2022
Electronic Engineer Intern *Hong Kong*

- Verified and validated a portable medical imaging device through testbench characterization and clinical trial, confirming compliance with design specifications.
- Proposed an alternative constant current source design to meet a lower load-regulation requirement.

LEADERSHIP EXPERIENCE

Deputy Teaching Assistant Coordinator Sep 2025 – Jul 2026
Electronic and Computer Engineering Dept., HKUST

- Plan and organize departmental programs and events for Graduate Teaching Assistants; provide peer mentoring.
- Host the monthly ECE Future Leaders PG Seminar Series, a platform for graduate students to expand knowledge within and outside their fields.

ECE Student Ambassador Sep 2023 – Jun 2024
Electronic and Computer Engineering Dept., HKUST

- Advised junior students on academic and non-academic matters; supported departmental outreach activities.

HONORS AND AWARDS

First Runner-Up — ECE Best Final Year Project/Thesis Award Jun 2024
Electronic and Computer Engineering Dept., HKUST

Dean's List Sep 2023
Electronic and Computer Engineering Dept., HKUST

Top 5 Innovative Solution — Sustainability Smart Campus Competition Nov 2021
The Hong Kong University of Science and Technology

SKILLS AND LANGUAGES

Circuit Design: Analog/Mixed-Signal CMOS IC Design, SAR ADC & Data Converters, mm-Wave RF Circuits (LNA), Power Management ICs, PCB Design, FPGA
CAD Tools: Cadence Virtuoso, Spectre, ADE, Synopsys, HSPICE, Altium Designer
Programming: MATLAB, Python, Verilog, Verilog-A
Languages: Korean (Native), English (Professional), Mandarin Chinese (Intermediate)